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Chapter 27: A Brief History of the Discrete Leviathan (Part II)—Understanding You (2012–2022)

Abstract

This chapter records how, between 2012 and 2022, platforms upgraded the behavioral recording and prediction that had already emerged in the "Collecting You" phase into sustained shaping of ever finer grain, higher frequency, and wider coverage—from "recording what you did" to "occupying and shaping your attention." The narrative unfolds along seven sections: the sharp fall of mobile data prices, the spread of handsets, and the shift from pull to push materially eased the bottleneck on individual-level training data (§1); EdgeRank rebuilt the visibility regime around "recent engagement," chronological feeds successively lost their default position across major platforms, and formats were compressed to ultra-short durations (§2); Skinner's intermittent reinforcement paradigm was engineered, through Eyal's Hook model, into observable platform practice (§3); TikTok replaced the social graph with the interest graph and converted creators into functions of metrics (§4); engagement-based ranking systematically amplified emotional and like-minded content under specific conditions, degrading the information ecosystem (§5); a decade of Western governance responses was never demonstrated to have changed the underlying architecture, leaving a systematic mismatch between the levels at which legal instruments operate and platform power operates, while China's new rules placed the recommendation mechanism itself under regulation earlier, with enforcement evidence pointing both ways (§6); and through format constraints, the industrialization of hooks, and engagement rewards, platforms reshaped users' expres-

sive habits from logic-dominant to rhetoric-dominant (§7). By the end of 2022, a system capable of predicting individual behavior and accordingly rearranging its perceived world had been fully built and was running at every layer; it locked itself onto the fragmentation of users' screen attention—an average of roughly 47 seconds between screen switches in information-worker samples (see the qualification in §8), converted creators into functions of watch-through rates, and reshaped users' expressive habits into hook openings, emotional closings, and rhetorical fragments standing in for chains of argument. The next step is no longer to predict and rank—but to generate and replace (§8, handing over to Chapter 28).

Keywords

algorithmic feeds; engagement optimization; variable rewards; interest graph; watch-through rate; format compression; attention fragmentation; information ecosystem degradation; dark patterns; from prediction to replacement

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§1 The Reversal of Material Conditions: Mobile Overtakes Desktop (2012–2016)

Cumulative investment in 3G/4G base stations drove the unit cost of mobile data down at an exponential pace. The unit price of mobile data dropped sharply over 2006–2012 (Jeanjean, 2014, 20-country panel data). Network-side per-MB costs fell 95% from 2G to 3G and a further 67% from 3G to 4G; users' combined per-MB spending fell 99% cumulatively between 2005 and 2013, and the threshold price of a usable smartphone dropped to $40 (Boston Consulting Group & Qualcomm, 2015). By around 2012, falling data prices and device thresholds had vastly expanded the range of affordable continuous connectivity—"always online" passed from the luxury of the few to the default of the many.

Traffic data record the behavioral turn. The annual average share of mobile devices (including tablets) in global web traffic rose from 8.5% in 2012 to 44.1% in 2016, and in October 2016 the monthly share overtook desktop for the first time (51.3% vs. 48.7%) (StatCounter, 2016; annual series via Visual Capitalist). In under four years, users' primary scene of access shifted from "sitting down at a computer" to "reaching for the phone at any moment." In parallel, control over information entry points underwent two landmark shifts: on December 4, 2009, Google extended personalized search from logged-in users to everyone—the same query could now return different results depending on a user's prior activity, and most users were unaware of it (Pariser, 2011); on July 1, 2013, Google shut down Reader, a widely used Really Simple Syndication (RSS) subscription service—the RSS protocol itself did not die; what truly exited was the major platforms' investment in users organizing information sources of their own choosing (Hölzle, 2013)—information was no longer something you went to find; it came to find you.

The structural consequence of these changes stacking up: the recordable behavioral events each user generated daily—clicks, dwells, swipes, skips—expanded by orders of magnitude. Recommender systems had been constrained by the abundance of individual-level training data. The collapse of prices, the spread of handsets, and the replacement of pull by push jointly eased this constraint. Platforms thereby obtained enough data to train predictive models of individual behavior—the collection infrastructure described in Chapter 26 kept running and expanding, and on top of it a new layer of capability began to grow: not merely recording what you did, but predicting what you were about to do, and accordingly rearranging everything you would see.

§2 The Product Reversal: From Timeline to Algorithmic Feed (2010–2018)

The price collapse and handset spread described in §1 supplied the material conditions, but material conditions do not by themselves change how information is distributed. What changed distribution was a series of product decisions: platforms moved the power to decide what users see from the users themselves into the hands of algorithms.

§2.1 Rebuilding the Visibility Regime (2010–2012)

This transfer was completed in three stages. The first was the 2010–2012 period of hardware carriers and the rebuilding of the visibility regime. Smartphones with high-resolution cameras (the 2010 iPhone 4 among them), together with the spread of 3G/4G mobile bandwidth, turned the phone into an always-carried video terminal, reducing to nearly zero the friction between pulling out the phone and starting to consume content. Platform-side product decisions clustered in the same period, and the one with the most structural consequence was Facebook's algorithmic reworking of the News Feed.

Before algorithms intervened, Facebook's feed was chronological: everything posted by one's friends was displayed in reverse-chronological order and with equal standing, and what a user saw depended on who had posted most recently. EdgeRank changed that premise. Media scholar Taina Bucher (University of Oslo), in her 2012 analysis, dissected EdgeRank's three publicly known variables: Affinity (the interaction history between users), Weight (the value assigned to interaction types, comments above likes), and Time Decay. The algorithm multiplied the three to compute for each piece of content a score for "deserving to be seen." Only content scoring high enough entered the user's default feed (Top News) (Bucher, 2012). Bucher ran a two-month tracking experiment with her own Facebook account: within the same time window, the real-time stream (Most Recent) displayed 280 items of content, while the default Top News showed only 45—roughly 12%–16% of content attained visibility (Bucher, 2012). That is to say, on Bucher's own account, the default feed showed her only a small fraction of what her friends had posted—visibility was rationed by algorithm, not a default right.

Bucher advanced a structural judgment: what EdgeRank constituted was not Foucauldian surveillance—not "the threat of being permanently seen"—but its opposite: "the threat of invisibility." On Facebook there is "not so much a 'threat of visibility' as there is a 'threat of invisibility' " that seems to govern the actions of its subjects (Bucher, 2012). The concrete shape of this mechanism: visibility was constructed as a reward for interaction rather than a default right. Non-participation was "punished" by the algorithm—by being made invisible. And by displaying highly interacted content, the platform manufactured the impression that "everyone is participating," installing participation as the norm—non-participation became deviance (Bucher, 2012). A circular logic operates here as well: to interact with content, the content must first be visible to you; interaction raises your Affinity score, making more content visible to you in the future. The popular become more popular; the ignored vanish more completely.

In February 2011, Facebook further changed the default setting of Most Recent from "all friends and pages" to "the friends and pages you interact with most"—a change made without notifying users. Users believed they were seeing the real-time full stream; in fact they were already reading an undisclosed filtered stream (Bucher, 2012). Birkbak and Carlsen (Aalborg University), in a 2016 analysis published in *Computational Culture*, pushed the point further: EdgeRank is not merely a ranking tool but a "self-justifying ordering"—it packaged algorithmic ranking as a response to "information overload," as if the algorithm were offering a service rather

than exercising power. Yet its core principle is unambiguous: "the world of Edgerank is ordered according to recent engagement" (Birkbak & Carlsen, 2016). In this world, "a post by a Facebook user with no friends does not exist in the world of Edgerank," and an inactive friendship is "hardly a relation at all." The standard for deserving to be seen was no longer the truth, civic value, or representativeness of content, but recency of engagement (Birkbak & Carlsen, 2016).

Other product decisions of the same period reinforced the turn. In 2012, YouTube adjusted its discovery mechanisms for search and suggested videos, shifting from rewarding clicks toward greater emphasis on viewers' actual watch time; the company subsequently reported increased watch time across the site (Meyerson, 2012). Facebook acquired Instagram—a company with merely 13 employees and roughly 30 million users—for $1 billion (April 9, 2012; Facebook, 2012; headcount and user figures per contemporaneous press reports), folding a potential competitor into the walled garden. Also in 2012, AlexNet won the ImageNet competition with a 15.3% top-5 error rate against 26.2% for the runner-up, and deep learning passed from laboratory breakthrough to deployable engineering infrastructure (Krizhevsky, Sutskever & Hinton, 2012). These events were not isolated: EdgeRank rebuilt the visibility regime, YouTube rebuilt recommendation priorities, the Instagram acquisition sealed the alternatives outside the wall, and AlexNet supplied the technical substrate that would make all of this indefinitely more refined.

§2.2 Format Compression and Real-Time Signals (2013–2015)

In the 2013–2015 data-engineering period, streaming big-data architectures (Kafka/Flink) matured, and platforms acquired a near-real-time streaming capacity to register fine-grained behavioral signals such as dwelling and skipping and adjust recommendations accordingly. Media formats themselves were compressed in the same period: Vine (January 2013), under Twitter, limited video to six seconds; Instagram promptly answered with fifteen-second video. These format limits were not technical bottlenecks but product design: ultra-short duration forced content to deliver its emotional strike within seconds—which is precisely what algorithms most easily quantify and rank. Watch time and completion thereby became core recommendation signals—liking costs 0.1 seconds of muscle memory, whereas watching through means the user has surrendered time that cannot be taken back. The algorithm thereby gained a preference signal more reliable than any explicit feedback: the user's actually spent time.

§2.3 From Real-Time to Right-Time: The Full Retreat of the Chronological Feed (2016–2018)

From around 2016, the major platforms one after another made algorithmic ranking the default entry—the chronological entry still existed, but it retreated from default to an option requiring deliberate switching. In February 2016, Twitter announced a reworking of its signature real-time reverse-chronological stream: algorithmically selected "best Tweets" would now be shown first—chronology receded behind the algorithm. A month later, in March 2016, Instagram followed, announcing that the feed would soon be "ordered to show the moments we believe you will care about the most" (Instagram, 2016). In November 2017, even Snapchat—whose core experience was the ephemerality of the instant—reworked its friend ordering: the new dynamic Friends page arranged friends according to how users communicate with them, which the company called a "more sophisticated Best Friends algorithm" (Snap Inc., 2017). Media scholar Bucher (2020) conceptualized this shift as an institutional transition from the "real-time web" to the "right-time web." She noted that the temporal regime of the broadcast era rested on what Paddy Scannell called "common public time"—television and

radio made audiences feel they shared one time frame. The algorithmic feed established the opposite principle: instead of bringing everyone into the same time frame simultaneously, it assigned each individual a personalized temporal regime calibrated by relevance and timing. Bucher's core judgment is that algorithmic media hinge on "producing and acting on relevance, timing, and personalized time regimes," reviving the classical Greek notion of kairos, "understood as an opportune time, timeliness, or indeed, 'right-time' " (Bucher, 2020). This means platforms decide not only what users see but when they see it—the time axis itself passed from passive container into an instrument of distribution power.

In the same period, deep neural networks entered recommender systems across the board. YouTube's recommendation architecture was rewritten wholesale as a two-stage deep neural model (candidate generation + ranking), and the ranking layer's objective function was precisely predicted watch time rather than click-through rate—because "ranking by click-through rate often promotes deceptive videos that the user does not complete ('clickbait') whereas watch time better captures engagement" (Covington, Adams & Sargin, 2016). Two engineering details deserve record. The first is the "golden 3 seconds": videos are first released to a small pool of a few hundred active users, and the abandonment rate over the opening seconds already enters the algorithm's evaluation—industry practice treats the first three seconds as decisive, and videos swiped away in the first three seconds are heavily down-ranked—creators are thereby forced to open with shock imagery or suspense. The second is the seamless loop: fifteen-second short videos link ending to beginning, so that before the user can react the video has played a second time, sending the algorithm a signal that cumulative viewing time exceeds the video's own length and encouraging a further surge of distribution. These two designs locked user behavior and algorithmic incentive into a positive-feedback loop: unconscious dwelling was read as preference; preference was amplified into more of the same content; more of the same extended dwelling further.

§2.4 Three Generations of Media Forms and the Transfer of Distribution Power

Viewed through media form, the transformations covered in this chapter span three generations of user-generated content (UGC). In the blog era, users actively subscribed, actively clicked, and read paragraph by paragraph; control over the flow of information lay with the reader. In the microblog/Twitter era, the 140-character format limit and follow-graph-driven feeds sliced information, but users still knew what they were reading and why. In the short-video era, the algorithm-driven push stream bypasses the rational text-decoding regions and acts on emotion and attention directly as audiovisual flow—users no longer "look for content"; they are "captured by content." The path of distribution power across these three generations is clear: editors → keywords → social relationships → behavioral prediction models. Each step transferred more of the power to decide from the user to the platform.

§3 The Engineering Deployment of Variable Rewards (2014–2020)

The product reversal described in §2 solved the question of what users see, but not why they keep coming back. The engineering foundation of user stickiness is variable-ratio reinforcement—the number of responses required for a reward is unpredictable, and schedules of this kind, in experiments since Skinner, produce high-frequency behavior that is hard to extinguish; the slot machine is the paradigm case. Eyal's framework extends "unpredictability" further, to the content and timing of rewards.

In 2014, Silicon Valley ad-tech entrepreneur Nir Eyal converted this principle into an executable product-design manual. In *Hooked: How to Build Habit-Forming Products* (Portfolio/Penguin, 2014) he proposed the four-step

loop: Trigger → Action → Variable Reward → Investment. The core lies in the third step: Eyal distinguished three kinds of variable rewards—rewards of the tribe (likes, comments, follows), rewards of the hunt (the next novel item in an endless scroll), and rewards of the self (completing a progress bar, leveling up)—and noted that the most habit-forming products tend to invoke several kinds at once. The book circulated widely in Silicon Valley product circles after publication (Eyal, 2014).

Productized practice of this mechanism was already observable in the same period. Three years later, on CBS News' *60 Minutes* segment "Brain Hacking" (aired April 9, 2017), former Google product manager Tristan Harris and neuroscience entrepreneur Ramsay Brown (cofounder of Dopamine Labs) described the machinery from two angles. Harris likened the smartphone directly to a slot machine: "Every time I check my phone, I'm playing the slot machine to see, 'What did I get?' This is one way to hijack people's minds and create a habit, to form a habit" (CBS, 2017). Brown described the concrete engineering: when an Instagram user receives 30 likes, the platform does not notify all at once; it withholds a portion and releases them in a concentrated burst when the algorithm judges the user may be about to leave—"They're holding some of them back for you to let you know later in a big burst" (CBS, 2017). More telling was Brown's subsequent explanation: "There's some algorithm somewhere that predicted, hey, for this user right now who is experimental subject 79B3 in experiment 231, we think we can see an improvement in his behavior if you give it to him in this burst instead of that burst" (CBS, 2017). In Brown's telling, every user is being treated as one sample in a continuously running experiment.

Harris further pointed to the asymmetry of compute between user and platform: "your telephone in the 1970s didn't have a thousand engineers on the other side of the telephone who were redesigning it … every day to be more and more persuasive. That was not true in the 1970s" (CBS, 2017). When thousands of engineers are dedicated to making a product harder to put down, while the user confronts it alone with bare willpower, the balance of the contest tilts heavily toward the side of engineering investment. Harris called this "a race to the bottom of the brainstem, where if I go lower on the brainstem to get you … using my product, I win" (CBS, 2017). Competition among platforms is therefore not a competition over user experience, but over who can more effectively bypass users' conscious judgment and trigger instinctive responses directly.

This competition operated not only between platforms but inside them. Harris, in an interview with Hari (2022), described its institutional mechanism: an engineer proposes a small change that would help users' attention; about three weeks later, a manager reviewing the dashboards asks for it to be rolled back—because the "time spent" metric has dropped. Harris's summary of this structure: "Their business model is screen time, not life time" (Harris, as quoted in Hari, 2022). This is not a conspiracy theory but the inevitable result of the incentive structure: a single improvement that causes an engagement metric to decline is liable to be rolled back in the organization's routine dashboard review.

§4 TikTok: The Interest Graph Replaces the Social Graph (2016–2022)

The product reversal and variable-reward deployment described in §2–§3 shared one premise: they were still organized around the social graph. Facebook's EdgeRank replaced chronology, but what it filtered was still the friends and pages the user had followed; YouTube's recommender turned to watch time—subscriptions and follows remained an important gateway, but recommendation increasingly depended on watch history. What a user saw was still largely determined by "whom you know" or "whom you subscribe to." Under this architec-

ture, creators retained a layer of "community buffer"—their audience was a fan community that had actively subscribed, not anonymous traffic allocated purely by algorithm.

§4.1 Architectural Innovation: Cold Start from Behavioral Signals

TikTok's structural innovation was to cancel this premise. ByteDance started in 2012 with the news aggregator Toutiao, launched the short-video app A.me in September 2016 and renamed it Douyin that December (Brennan, 2020), named its international version TikTok in 2017, and in August 2018 merged the previously acquired Musical.ly into TikTok (Brennan, 2020).

Business journalist Matthew Brennan, in his ByteDance study *Attention Factory* (2020), notes that the recommendation engine "was at the very heart of TikTok. It was the key to understanding the success of the app and its parent company, ByteDance" (Brennan, 2020). Unlike Facebook and YouTube, ByteDance made the recommendation system its only distribution architecture from the company's founding, rather than layering an algorithm on top of an existing social graph. Brennan calls it "the earliest Chinese internet company to go 'all in' " on the then-nascent technology of recommendation engines (Brennan, 2020). TikTok's For You Page (FYP) requires no following to start working: from the first second a new user opens the app, she is inside the algorithmic feed, and the cold start is driven primarily by behavioral signals—watch duration, completion, swipe velocity, rewinds, shares. The weight of "whom you know" in the distribution algorithm approaches zero; "what you did" becomes the primary determinant. That is the concrete meaning of "the interest graph replaces the social graph." The transformation thoroughly reconstructed the relation between creators and audience: their audience was no longer a community of active subscribers but anonymous traffic allocated predominantly by algorithm.

Media scholar John Jordan (DePauw University) drew out the core difference: "YouTube was built to serve horizontal video windows to people who chose from a selection on a computer. TikTok serves one vertical video at a time on a smartphone" (Jordan, 2024). ByteDance founder Zhang Yiming's stated objective revealed a deeper ambition—"to eliminate the need for search across all corporate properties" (Zhang Yiming, as quoted in Jordan, 2024)—standing in fundamental opposition to Google's mission of "organizing the world's information": Google presumes the user has active intent, and the platform's job is to help you find it; ByteDance presumes the user has no active intent, and the platform's job is to push it to you before you know what you want.

The success of this architecture forced every competitor to follow. Instagram launched the recommendation-driven Reels short-video feature in 2020 (Chayka, 2024); YouTube launched the early beta of Shorts in India on September 14, 2020, then extended it globally (Jaffe, 2020); Twitter, after Elon Musk's acquisition, introduced the "For You" column in 2022 (it became the default in January 2023; Chayka, 2024). Chayka (2024) records this competitive imitation: TikTok was "reaching more than 1.5 billion users in less than five years," and "its competitors, struggling to catch up, have followed suit into algorithmification" (Chayka, 2024). This means the architecture described in §4.1 is not a TikTok peculiarity but had become the distribution architecture that every major platform has raced to adopt.

§4.2 The Distribution Funnel and the Alienation of Creators

This architecture's impact on the supply side ran deeper than on the consumption side. Penn State political scientist Kevin Munger, in *The YouTube Apparatus* (Cambridge Elements, 2024), advances a central thesis: in

platform-mediated video distribution, producers primarily consume audience metrics rather than content, and therefore "we should thus expect larger effects on the producers than on the consumers" (Munger, 2024). On platforms like YouTube, which still preserve subscription and community loyalty, creators at least retain a fan community as buffer; the audience consists of community members who chose actively, not anonymous traffic assigned by algorithm. Yet even under these conditions, "the immediate and highly specific and quantified feedback that creators of online content receive enables (and almost forces) them to figure out what the audience wants" (Munger, 2024). Lewis's (2018) research adds that audiences exert a form of accountability pressure on creators through engagement metrics, and this "can push influencers toward more extremist viewpoints—in other words, radicalization can happen in reverse" (Lewis, 2018). Munger's judgment transforms the very concept of "creator": "Although YouTube encourages these independent YouTubers to call themselves 'Creators,' we should instead think of them as Creations" (Munger, 2024).

TikTok pushed this logic to the extreme. When distribution is driven primarily by behavioral signals and no longer rests on subscription, no stable community bond remains between creator and audience. Brennan (2020) describes the distribution architecture in detail: each uploaded video first passes automated review (computer-vision analysis, keyword tagging, duplicate detection), then is released to "a small pool of a few hundred active users"; the algorithm evaluates performance metrics—complete views, likes, comments, average play length, shares—and well-performing videos advance to the next tier, facing thousands and then millions of users (Brennan, 2020). The underground-industry operators Brennan quotes treat watch completion as the single most important metric—"We heard the 'watch completion rate' was most important" (Brennan, 2020). A creator's historical follower count has almost no effect on a single video's distribution. The result: the creator no longer faces a recognizable community but an opaque distribution funnel, with audiences allocated as anonymous traffic. This structure converts creators into functions of metrics more thoroughly than YouTube: since completion is the funnel's core metric, the first three seconds must manufacture suspense to suppress swiping, the ending must loop back to manufacture a signal of cumulative viewing beyond the video's own length, and structure must obey the upgrade logic of "test pool → small pool → large pool." The creator is not expressing; she is sending results to a continuously running A/B test.

§4.3 Economic Structure and Legal Recognition

Economic structure reinforced the asymmetry. Venture capitalist Chris Dixon (a16z partner), in *Read Write Own* (Random House, 2024), calculates that the advertising take rate of Facebook, Instagram, TikTok, and Twitter runs to roughly 99%: the platform captures nearly all advertising revenue, and creators' direct compensation amounts to less than 1%. YouTube is the exception, with a take rate around 45%, returning the remaining 55% to creators (Dixon, 2024). TikTok later established a "creator fund," but the fund is a fixed-total pool rather than revenue sharing—YouTuber Hank Green noted: "When TikTok becomes more successful, creators make less per view" (Green, as quoted in Dixon, 2024). This means creators produce content for the platform, are trained by metrics, and supply training data to the algorithm while receiving almost nothing from the platform's direct advertising split. Expression is compressed into algorithm-recognizable formats, direct monetary returns on labor are pressed to a minimum, and "visibility" becomes the creator's principal capital for converting outside opportunities within this distribution structure. Legal scholars Adams and Grosse Ruse-Khan (University of Cambridge), in a 2020 analysis in the *International Journal of Law and Information Technology*, provide the theoretical frame for this structure: digital platforms such as YouTube, Facebook, and Instagram have monopolized an ever-larger share of public attention, on which individuals increasingly depend as outlets for their

creative work; the intellectual-property system enables firms to enclose and appropriate the products of that activity—for "free," or at token rates—with the result that the interests of creators and the public are subordinated to the demands of capital accumulation (Adams & Grosse Ruse-Khan, 2020). Creative work is often presented as the antithesis of capital's work—autonomy, self-expression—yet is increasingly harnessed to serve capital accumulation.

Legal recognition of this architecture has begun to take shape. A 2026 online commentary in the Harvard Law School's JOLT Digest analyzed the enforcement lawsuits brought against TikTok by fourteen states in 2024. The plaintiffs alleged that its recommendation system is "aimed solely at maximizing users' engagement with the platform, without any concern for the subject matter of its recommendations"; and, at the motion-to-dismiss stage—taking the complaint's facts as true—some courts held that such purely behavioral-signal-driven recommendation does not constitute editorial judgment protected by the First Amendment—in Zhao's formulation, "No human programmer at TikTok is 'disfavoring' any content; the algorithm aims only at user engagement" (Zhao, 2026). This is a procedural ruling, not a final factual finding on how the system actually operates. It stands in contrast with the editorial judgment of traditional media: distribution here is based on no judgment about the content itself, but is primarily a response function to user-behavior data.

Seen along the evolution of distribution power, TikTok is the purest form of a fourth-generation architecture. The first generation was portal editors deciding the order of content; the second, users actively searching by keyword; the third, social-relationship-driven feeds; the fourth, a distribution model led by behavioral prediction. In the first three, the user made an active choice at some link in the chain (clicking a link, typing a keyword, following someone). In the fourth, the user's only "choice" is whether to keep watching. The swipe itself is data; the dwell itself is a vote. The user no longer "looks for content"; content captures her precisely.

§5 Engagement Optimization and the Content Ecosystem (2018–2022)

The architectural turn described in §4 explains how platforms distribute content, but not what happens to the ecosystem when engagement becomes the sole optimization target. The answer emerged gradually after 2018: platforms were not transmitting information; they were systematically amplifying content capable of triggering emotional response.

§5.1 Engagement-Based Ranking and Information Distortion

§2 described how EdgeRank redefined what deserves to be seen with "recent engagement" as the sole criterion. By 2018, that logic had evolved, on all major platforms, into engagement-ranking systems more complex in form but identical in essence. The consequence is not confined to any one company; it is a structural tendency of engagement ranking under specific conditions.

Facebook's turning point came in early 2018. The platform retooled its feed algorithm around "Meaningful Social Interaction" (MSI), increasing the weight of content actively engaged with through comments or likes. One component, "Downstream MSI," sought to predict what content would go viral and boost its prominence. The tens of thousands of internal pages leaked to the U.S. Securities and Exchange Commission (SEC) and Congress in October 2021 by former Facebook data scientist Frances Haugen showed that this change increased the spread of angry content. A December 2019 internal document stated: "Unlike communication with close friends and family, virality is something new we have introduced to many ecosystems…and it occurs because

we intentionally encourage it for business reasons." The same document conceded: "Our ranking systems have specific separate predictions for not just what you would engage with, but what we think you may pass along so that others may engage with. Unfortunately, research has shown how outrage and misinformation are more likely to be viral." Its recommendation was to "stop magnifying harmful content by giving it unnatural distribution" (Harari, 2024). The leaked internal research also contained more concrete damage figures: as the *Wall Street Journal*'s Facebook Files series revealed, Instagram's internal research showed that among teens who reported suicidal thoughts, 13% of British users and 6% of American users traced the desire to kill themselves to Instagram; among girls who reported body-image concerns, one in three said Instagram made them feel worse—a 2019 internal slide summarizing research on teen girls who experience these issues read "We make body image issues worse for one in three teen girls"; and CBS's *60 Minutes*, reporting on the same body of internal research, noted that 17% of teen girls said Instagram made eating disorders worse (Wall Street Journal, 2021; CBS News, 2021; the denominators are those who reported each issue).

But this was not only Facebook's problem. In 2021, the algorithm team at Twitter (now X) ran internally the political-content amplification study—the most comprehensive audit of a recommender system's political effects yet conducted, led by the platform itself (Huszár et al., 2022; co-author Luca Belli was himself an algorithm researcher at the platform—Belli, 2025). Through a long-running randomized experiment (since 2016, roughly 1% of global users had been randomly retained on a reverse-chronological feed as a control group), the team measured the algorithmic amplification of legislators' content. The findings: Twitter's algorithm systematically amplified the visibility of legislators' political content (amplification ratios were positive in most countries studied and exceeded 200% in some; a 200% amplification ratio corresponds to roughly three times the reach of chronological ranking); across the seven countries' mainstream parties, in six of the seven the mainstream right enjoyed higher algorithmic amplification than the mainstream left, and the analysis of the US media landscape likewise showed right-leaning outlets receiving higher amplification; the study found no evidence, however, that far-left or far-right groups were amplified more than moderate ones. This finding accords with the conditional mechanism below: the structural bias of engagement ranking is not any one product team's error but a property of the engagement metric under specific conditions—political content more readily triggers shares and comments, and where the condition "more extreme users engage more" holds, engagement ranking amplifies its excess exposure (see the analysis of conditions in the Germano passage at the end of this section).

In 2023, a large-scale randomized experiment conducted by researchers from seventeen universities in collaboration with Meta provided the most direct causal evidence. Guess et al., in a study published in *Science*, randomly assigned a group of US users to a reverse-chronological feed (rather than the default algorithmic feed) for three months. The results: the algorithmic feed significantly increased the time users spent on the two platforms and their volume of interaction; on Facebook's ideological-classification measures, it increased the share of exposure to "like-minded" sources (53.7% vs. 48.1%) and decreased the share of moderate/mixed sources (22.6% vs. 30.9%). Exposure to political and untrustworthy content, however, increased after the switch to the reverse-chronological feed (Guess et al., 2023)—the algorithmic feed's effect on "which sources you see" varies by platform and metric. Independent field experiments supply causal evidence from the other side: on Bluesky, 2,000 users were randomly assigned for eight weeks to one of three feeds—engagement-based ranking, reverse-chronological, or a ranker down-weighting extreme users; engagement-based ranking significantly amplified moral outrage and toxic political content (moral outrage up to +79.2% after the election) and raised perceived

partisan animosity (Brady et al., 2026); on X, down-ranking hostile content likewise reduced out-party animosity (Piccardi et al., 2025). Engagement ranking is therefore not a neutral instrument of transmission: it systematically amplifies emotional and moralized content and systematically concentrates like-minded sources, but its net effect on untrustworthy sources depends on platform and metric—nor did users' attitudes change significantly over the three months of the Guess experiment, whose long-run effects lie beyond what that experiment can determine.

Germano, Gómez, and Sobbrio (Universitat Pompeu Fabra / Università Tor Vergata), in a 2025 working paper, provide a unified theoretical account of these scattered observations. Using the MSI update as a natural experiment and difference-in-differences estimates on representative survey data from Italy and the United States, they find that after the update, users who relied on Facebook for political news were significantly more likely to hold extreme ideological positions and to exhibit stronger affective polarization. Their theoretical model reveals the general form of the mechanism: when the ranking algorithm gives greater weight to likes/shares (highlights)—the parameter η—users with more extreme beliefs, who are naturally more prone to like and share congenial content, exert disproportionate influence on the ranking → the algorithm ranks such extreme content higher → moderate and accurate content is crowded out. The mechanism's 2022 working-paper version was titled "crowding out the truth?" The crucial point: the model shows that when users with more extreme beliefs are also more prone to like and share, raising the ranking weight of such interactions (η) magnifies their influence on visibility and worsens information distortion and polarization—this is a conditional mechanism, and its condition is supported by the relevant empirical literature; in the flat case, where the propensity to highlight does not rise with extremity, raising η simultaneously decreases both polarization and misinformation. It is therefore not an unconditional law of every engagement-ranking system, but a structural tendency that holds when the condition "extreme users engage more" is satisfied. A further set of independent evidence comes from González-Bailón et al. (2023) in *Science*: on a dataset of 208 million US adult active Facebook users, they confirmed that ideological segregation rises significantly at the algorithmically curated exposure stage on Facebook. Braghieri, Levy, and Makarin (2022), exploiting the natural experiment of Facebook's staggered rollout across US colleges, provide causal evidence at the mental-health level: Facebook's arrival on campus significantly worsened students' mental health and increased reports of academic impairment from psychological conditions, the mechanism being the platform's fostering of unfavorable social comparison.

§5.2 The Weaponization of Open Data (2018)

Another consequence of the engagement-optimization model surfaced the same year. In March 2018, the British political consultancy Cambridge Analytica was revealed to have harvested the personal information of a vast number of Facebook users without authorization—per Facebook's April 2018 disclosure, up to roughly 87 million users were affected (Facebook, 2018). The mechanism: in 2013, Cambridge data scientist Aleksandr Kogan developed a personality-quiz app, "This Is Your Digital Life," recruiting participants via Amazon Mechanical Turk at one dollar apiece; once a user completed the quiz on Facebook, the app—through Facebook's platform interfaces—collected not only the user's own data but the personal information of her entire friend network, most of whom had never interacted with the app (Kaiser, 2019). Cambridge Analytica used these data to build psychographic profiles, inferring personality traits from users' Facebook activity and then showing tailored messages to different voters during the 2016 US presidential election (Kaiser, 2019). What the episode revealed was not merely a data breach but a structural risk: platforms expanded behavioral-data collection to improve prediction while also opening interfaces to third-party developers, allowing data on users and their friend net-

works to leave the platform—and once data left, it could be weaponized. In July 2019 the US Federal Trade Commission (FTC) fined Facebook $5 billion—see §6.2.

§5.3 Ecosystem Degradation

The consequences of engagement ranking appeared not only as intensified political polarization but as the structural degradation of the entire information ecosystem. This degradation occurred simultaneously in developed and developing countries, though by somewhat different mechanisms: in developed countries it chiefly took the form of the collapse of professional journalism and the flattening of culture; in developing countries, the direct amplification of violence and ethnic hatred. The mechanism common to both is one and the same: when platforms allocate visibility primarily by engagement, content that triggers fast emotional response more readily gains outsized distribution and can crowd out content that demands cognitive effort.

In the United States, more than a quarter of newspapers have closed since 2004, most of them local, creating vast "news deserts"—hundreds of communities lost their only local news source (Abernathy, 2020, as cited in Lawrence & Napoli, 2024). Surviving newsrooms faced a double squeeze: on the revenue side, dependence on platform distribution; on the content side, the pull of social-media performance metrics. The work of communication scholars Fürst (2020), Lamot (2022), and Lischka & Garz (2021) jointly shows that the "tabloidization" of news—sensational headlines and soft news displacing original reporting and hard news—correlates directly with newsrooms' "obsession" with social-media performance (as cited in Lawrence & Napoli, 2024). The degradation is gradual enough that audiences may not perceive the decline in quality. Shin (2024, in Lawrence & Napoli, 2024) concludes: the decline of news quality inside the mainstream media may be more serious than misinformation itself.

One internal YouTube experiment provides evidence in the reverse direction. Jonathan Stray (2020) recorded what happened when YouTube lowered the visibility of low-quality "tabloid" content: watch time initially fell—precisely what the engineering team feared—but within three months, users' time on high-quality videos not only recovered but grew (as cited in Lawrence & Napoli, 2024). The experiment reveals a key fact: content degradation is not the natural preference of "users liking junk," but path dependence created by algorithmic recommendation—once the algorithm's prioritization of low-quality content was removed, user behavior reversed within a short period.

Culture critic Kyle Chayka, in *Filterworld* (2024), condenses the process into "the flattening of culture": the recommendation algorithm does not create diversity; it pushes all culture toward the lowest common denominator. Jingjing Zhang, an Indiana University professor who studies recommendation algorithms, found in an experiment that over time the system will "provide less diverse recommendations" and ultimately will "provide similar items to everyone" (Zhang, on the Planet Money podcast, as cited in Chayka, 2024). This is not the mirror image of political polarization—political content divides audiences through controversy; cultural content aggregates audiences through safety—but it is the other face of the same optimization function: whatever keeps more people watching longer gets recommended to more people. The result is the systematic suppression of independent, local, and experimental cultural expression.

The same mechanism produced more extreme consequences in the weak institutional environments of the Global South. Myanmar is the starkest case. Historian Yuval Noah Harari (Hebrew University), in *Nexus* (2024), documents Facebook's role there in detail. From 2012 to 2015, internet adoption in Myanmar exploded from 0.5% to

40%, mostly through cheap smartphones (Fisher, 2022). Through zero-rating, Facebook made its app free to use, so that for many Burmese, Facebook was the internet itself (Fisher, 2022). In 2014, Facebook employed only one Burmese-language content moderator for the entire country (Harari, 2024). In 2016, Facebook launched Instant Articles in Myanmar, paying news outlets by engagement—clicks and views—with no regard for the truth of content. Harari cites a 2021 study: in 2015, before the program was launched, six of Myanmar's top ten Facebook pages belonged to legitimate media; by 2017 only two remained; by 2018 all ten pages were fake-news and clickbait sites. The study concluded that "clickbait actors cropped up in Myanmar overnight. With the right recipe for producing engaging and evocative content, they could generate thousands of US dollars a month in ad revenue, or ten times the average monthly salary—paid to them directly by Facebook" (Harari, 2024).

The same structure reproduced itself independently in Sri Lanka. When *New York Times* journalist Max Fisher arrived in early 2018, "whole villages, as if suddenly possessed, had formed into mobs, ransacking and burning their neighbors' homes" (Fisher, 2022). The trigger was an eighteen-second phone video circulated through Facebook groups, showing a Muslim restaurant owner confessing under duress to "poisoning." As in Myanmar, Facebook had achieved vertical integration of the information infrastructure through zero-rating; local human-rights organizations repeatedly reported hate speech, and the platform ignored them. In the end the government was forced to block Facebook to stop the spread of the riots. Fisher, relaying *The Economist*'s analysis of Twitter's 2016 algorithmic redesign, sums up: the engagement-optimizing algorithm "appears to reward inflammatory language and outlandish claims" (Fisher, 2022).

§5.4 Internal Knowledge, External Denial

Facebook's disregard of internal warnings was not confined to the Global South. Instagram's internal documents show employees had a clear understanding of their own product. In November 2025, the plaintiffs in the US federal multidistrict litigation (MDL 3047), in their opposition to the defendants' motions for summary judgment, quoted internal Meta materials showing that Instagram employees internally described themselves as "basically pushers"—"We're basically pushers"—and that internal documents recorded that teens were becoming addicted even though it made them feel bad, that Instagram was addictive, and that time spent on the platform was negatively affecting their mental health (MDL 3047 court filing, submitted November 21, 2025; internal evidence invoked by the plaintiffs, not a judicial finding). This is corroborated by the internal Facebook research leaked by Haugen in 2021: the platform internally knew its addiction-by-design harmed teenagers, and did not stop.

Adam Mosseri, then Facebook's vice president in charge of the News Feed, offered a simplified version of the logic at a Facebook conference: "If they do these interactions a bit more, that's an indication that we're creating some value. Over the long run, more and more people will use Facebook, they'll spend more and more time on it, and that will be good for them, it'll be good for Facebook, and it'll be good for publishers" (as quoted in Fisher, 2022). YouTube's algorithm chief Cristos Goodrow said much the same: "When users spend more of their valuable time watching YouTube videos, they must perforce be happier with those videos" (as quoted in Fisher, 2022). Fisher's comment on this: "It was a strange assumption. People routinely act against their self-interests. Whole worlds of expertise organize around the understanding that our impulses can overpower us, usually to our detriment" (Fisher, 2022).

§5.5 A Systemic Condition, Not Individual Pathology

Hera Hyeonseo Lee, a doctoral student in sociology at Binghamton University and research associate at the AI Now Institute, provides a structural explanation in a 2026 paper in *New Media & Society*. Using the concept of "brain rot," she argues that attention fragmentation and cognitive decline are "not a glitch…but a structural externality of attention assetization" (Lee, 2026). Fragmented attention is productive within the platform's economic logic: a user deeply focused on a two-hour movie generates one data point, while the same user scrolling short videos for two hours generates hundreds of discrete behavioral signals. The latter is worth far more to the platform's algorithmic training. Platforms are therefore not "distracting users' attention"; they are economically incentivizing content forms that can slice attention into more discrete signals. Lee's conclusion: brain rot is a systemic condition rather than an individual pathology—not the result of a user's failure of self-discipline, but a structural product of the algorithmic economic model.

§6 Governance Responses (Cross-Cutting, 2012–2022)

§1–§5 described how, over a decade, platforms built an architecture of behavior prediction and content distribution centered on engagement. This section asks the symmetric question: what did governance do in those ten years? The answer: a great deal—but by the end of 2022, not a single intervention had been demonstrated to have changed the underlying architecture (the existence of rules is not the effect of rules, see the end of this section). The reason is not lack of will but that governance instruments were repeatedly deployed at the layers of data disclosure and after-the-fact accountability, while recommendation objectives and ranking mechanisms themselves remained outside mandatory review—a systematic gap between the layer at which the tools operate and the layer at which platform power operates.

§6.1 The Interface Bypass of Consent (GDPR and Deceptive Patterns)

On May 25, 2018, the EU's General Data Protection Regulation (GDPR) became applicable. GDPR requires that consent be "freely given, specific, informed and unambiguous" (Art. 4(11)). This was the most ambitious privacy legislation since the 1990s data-protection directives, covering in principle every platform operating in the EU. But between the legal text and the user's lived experience lay a structural gap: the law specified the conditions of consent—including that withdrawal be as easy as giving—yet prescribed no uniform button layout, wording, or workflow for consent interfaces.

Platforms exploited this gap with systematic interface bypass. Barely a month after GDPR took effect, the Norwegian Consumer Council (Forbrukerrådet) published the investigation *Deceived by Design* (June 27, 2018), finding that Facebook and Google used deceptive patterns in their privacy-settings interfaces to nudge users away from privacy-friendly options. The concrete method: accepting privacy-intrusive settings takes one tap of "accept and continue," while refusal has no matching button—the user must click the vaguely labeled "manage data settings" and then operate a switch of unclear direction (Brignull, 2023, citing Forbrukerrådet, 2018).

This was no isolated case. HCI researcher Nouwens et al. (2020) showed in an experiment with 40 participants that merely removing the "reject all" button from the first screen of a cookie-consent popup could raise consent rates by up to 23 percentage points (Nouwens et al., 2020). Chile's consumer-protection agency SERNAC, in a 2022 field experiment with 72,208 valid unique visitors, further quantified the power of defaults: across otherwise identical pop-ups differing only in their defaults, 86.72% of users rejected additional cookies when

they were unselected by default (opt-in), compared with only 5.60% when they were preselected (opt-out)—a gap of 81.12 percentage points; relative to a strong-dark-pattern control, the report separately estimated that the opt-in default raised rejection by about 86 percentage points (SERNAC, 2022). Harry Brignull, the namer of deceptive patterns, summarized the structural predicament: cookie-consent dialogs often demand so much effort to exit that users grow exhausted until they give up (Brignull, 2023). GDPR's legal intent was systematically dissolved at the interface layer.

§6.2 America's Legal Vacuum (§230 and the FTC)

The governance gap on the American side was more fundamental. Section 230 of the 1996 Communications Decency Act gave online platforms dual protection: it immunized platforms from publisher liability for user-posted content, while allowing them to moderate and curate content without thereby losing that immunity.

Platform scholar Tarleton Gillespie (Cornell University / Microsoft Research) points out the consequence: platforms could simultaneously claim "I am not a publisher" to escape liability while continuously exercising publisher-like powers of ranking, recommending, deleting, and amplifying—he describes platform intervention as extending beyond moderating, deleting, and suspending to "privileging some content over others, in opaque ways" (Gillespie, 2018).

In this book's analytic framework, the engagement-optimization architecture described in §2–§5—algorithms amplifying angry content, MSI predicting virality, recommender systems driving creator extremity—is shielded by §230 to a large extent. §230 has explicit exceptions, and the absence of a federal statute aimed specifically at recommender algorithms does not mean product-design conduct lies beyond all federal law—the FTC pursued the platform's deceptive privacy representations under Section 5 of the FTC Act; but as for "the manner of algorithmic distribution" itself, a dedicated federal cause of action is indeed missing.

In July 2019, the FTC answered the Cambridge Analytica scandal with a $5 billion fine—the largest penalty ever imposed on any company for violating consumers' privacy, almost 20 times greater than the largest privacy or data security penalty ever imposed worldwide (FTC, 2019). But the settlement did not require Facebook to change the design of its recommendation algorithm, did not limit the scope of engagement optimization, and did not require third-party audits of algorithmic behavior. The settlement did require privacy-governance reforms and independent assessments—but all of those arrangements targeted "data handling"; none touched recommendation objectives or ranking mechanisms. The fine targeted the consequences of the data breach, not the architecture that produced it. By the congressional hearing of March 2021, Representative Lisa Blunt Rochester asked Zuckerberg directly whether he would commit to including deceptive patterns in his "better internet regulation" proposals. Zuckerberg's reply: "I'll think about it. My initial response is that I feel there are other areas that I think might be more urgently in need…" (Brignull, 2023, quoting the March 2021 hearing video). Two years after a $5 billion fine, the platform's chief executive was still publicly declining to commit to ending manipulative interface design.

§6.3 The Shift in the Evidentiary Landscape (Haugen and Molly Russell)

What truly changed the terrain was not law or fines but the leaking of evidence. In October 2021, former Facebook data scientist Frances Haugen submitted tens of thousands of internal pages to the US Securities and Exchange Commission and the *Wall Street Journal*, and testified before the Senate. Those documents—as §5

recounted—proved the company internally knew its product harmed teenagers and chose not to fix it. The institutional significance of the Haugen leak lies not in any single revelation but in its destruction of the premise of the "we didn't know" defense: after internal documents explicitly recorded that virality was deliberately encouraged for commercial reasons, that defense was no longer sustainable. But a leak is not a legal verdict—it changes the evidentiary landscape, not the legal rules.

A year later, the first causal chain in law was established in the United Kingdom. On September 30, 2022, London's senior coroner Andrew Walker delivered his findings on the death of 14-year-old Molly Russell. Molly died by self-harm in November 2017; in the six months before, she had saved, liked, or shared a total of 16,300 pieces of content on Instagram, of which 2,100 related to depression, self-harm, or suicide (inquest evidence)—much of it pushed by algorithm rather than actively sought. The coroner concluded: "It is likely that the above material viewed by Molly, already suffering with a depressive illness and vulnerable due to her age, affected her mental health in a negative way and contributed to her death in a more than minimal way" (Walker, 2022). The National Society for the Prevention of Cruelty to Children (NSPCC) called it the first ruling worldwide that content a tech company allowed and encouraged children to view had contributed to a child's death. But this remains a coroner's inquest, not a damages verdict—the causal chain was established; the compensation loop has not closed.

By the end of 2022, the net result of a decade of governance responses can be summarized as follows: a consent framework systematically bypassed by interface design (GDPR); a record fine requiring no change to product architecture (FTC, $5 billion); an internal-document leak that destroyed the "we didn't know" defense yet produced no legally binding constraint (Haugen); and a coroner's finding that established a causal chain for the first time yet produced no compensation (Molly Russell).

Among non-Western states, India banned TikTok and 58 other Chinese apps on national-security grounds in June 2020, and 200 million monthly active users went to zero overnight—but that was a geopolitical decision, not regulation of algorithmic design itself. The engagement-optimization architecture described in §1–§5—algorithmic ranking, variable rewards, behavior-signal-driven recommendation—only became a direct object of rules in force in the major Western jurisdictions with the EU's Digital Markets Act (DMA) and Digital Services Act (DSA): the DMA entered into force on November 1, 2022, followed by the DSA on November 16, 2022. Most DMA provisions became applicable in May 2023, and the DSA became fully applicable in February 2024; their enforcement belongs to the next chapter's period.

China moved earlier—Article 24 of the 2021 Personal Information Protection Law and the Internet Information Service Algorithmic Recommendation Management Provisions, in force since March 2022, directly regulate automated decision-making and algorithmic recommendation design, and the object of regulation includes the recommendation mechanism itself: Article 18 of the E-Commerce Law and Article 24 of the PIPL require that information-push services simultaneously offer options not targeted at personal characteristics; the July 2021 guiding opinions issued by seven government departments for food-delivery platforms directly intervened in assessment-algorithm parameters ("algorithmic moderation," loosened delivery deadlines) (State Administration for Market Regulation et al., 2021); the "Qinglang 2022 Comprehensive Algorithm Governance" campaign, launched in April 2022 and running to early December, organized algorithm-security self-inspection and rectification across enterprises, conducted on-site inspections of some internet companies, and pressed for algorithm filing—the first batch of domestic internet information service algorithm names and filing numbers was published in its wake (Cyberspace Administration of China, 2022; Xinhua News Agency, 2022). Under

Articles 16 and 17 of the Provisions, platforms must publish the basic principles of their algorithms and provide a convenient option to switch off algorithmic recommendation services—before the Provisions took effect in March 2022, several major apps had already rolled out user-facing switches for personalized recommendation (China City News, 2022). On the judicial side, the first successful "big-data price discrimination" lawsuit (Hu v. Shanghai Ctrip Business Consulting, first-instance ruling in July 2021, final judgment on December 31 of the same year) ordered the platform to refund the price difference and pay three times that amount in damages, and to add to its app an option to continue use without accepting the existing terms. Within this chapter's period, then, enforcement of the new rules was already observable; but the existence of rules is not the effect of rules—a study published in the same year the Provisions took effect noted problems of "difficulty in implementing the provisions and vague institutional design" that reduced its operability (Xiao & Shang, 2022); subsequent research has documented the doubts raised by "indeterminate content boundaries and the absence of an information-disclosure mechanism" in algorithm filing (Liu, 2025), as well as the difficulties facing both the filing system and transparency proposals under black-box conditions (Zhang, Yan & Zhang, 2024; cf. ch08B: the evolution of rules by itself cannot prove a sustained decline in narrative leverage—two-sided enforcement evidence likewise does not license reading the regulatory text as accomplished fact). This contrast does not overturn the section's main thesis, but it corrects its scope: the Western mismatch lies in the object of regulation excluding ranking mechanisms, whereas China's gap lies in the depth and transparency of enforcement.

In the Western jurisdictions, there is a systematic mismatch between the level at which governance instruments operate—the legality of data collection, after-the-fact fines for privacy violation, the public disclosure of evidence, the establishment of case-by-case causal chains—and the level at which platform power operates—interface defaults, recommendation ranking, the engagement-optimization objective function. The mismatch is not accidental oversight; it has a deeper structural cause: the prevailing governance framework rests on the "informed consent" paradigm, and that paradigm presupposes a decontextualized autonomous subject—one who can independently assess privacy costs, make rational trade-offs among hundreds of terms of service, and grant meaningful advance authorization for future uses of data.

Legal scholar Yeung (2016) argues that in algorithmic environments this presupposition fails systematically: the individual's decision environment is itself being continuously restructured by platforms—every algorithmic ranking change recorded in §2–§4 alters the user's information choice architecture in real time—so the stable decisional situation that "consent" presupposes does not exist. The interface bypass recorded in §6.1 is not merely technical design deception; it reveals a conceptual-level failure: when the choice environment is dynamic, personalized, opaque, and commercially driven, a governance framework built on static one-off consent cannot reach the question "in what manner you may present and amplify content." The law drew its line at "what data you may collect"; on the question "in what manner you may present and amplify content"—that is, the entire architecture recorded in §2–§5—it had not yet found a grip. In November 2022, the EU's DSA and DMA took effect in succession, for the first time reaching algorithmic transparency, recommendation-system design, and gatekeeper conduct at the legal level—but their subsequent application and enforcement belong to the next chapter's period.

§7 The Formatting of Expression: From Internalized Constraint to the Ascendancy of Rhetoric

The preceding six sections recorded what platforms did to users—recommending, ranking, rewarding, optimizing. But platforms did something more covert: they reshaped the forms in which content is published and

browsed—expression and attention were reshaped through those two forms, and ways of thinking came under pressure. Not through censorship, but through format.

§7.1 The Internalization of Constraint (2006–2013)

Twitter's 140-character limit was not a product-philosophy choice—it descended from the 160-character SMS cap Friedhelm Hillebrand proposed for the GSM standard in 1985, with Twitter reserving 20 characters for the username and leaving 140 for text (Hillebrand, 2010; Liao, Fu & Hale, 2015). A technical constraint of a telecom protocol became a literary form. Over seven years users internalized this accidental constraint as a habit of expression: short sentences, aphorisms, rhetorical fragments displacing chains of argument. When Twitter expanded to 280 characters in 2017, the expressive habit had become hard to revert in the short term—the culture had been deeply shaped by a technical constraint. Weibo followed the same path.

In this process a deep conversion occurred: "posting opinions" became "posting chatter." The physical space of 140 characters makes a complete, developed argument (premise–inference–conclusion) difficult to hold, but it can hold an emotional expression, a rhetorical posture, a slogan. The platform need not ban long-form—it need only let short-form win on engagement metrics, and long-form is marginalized in the visibility contest. The dominant form of expression shifted from logic-dominant to rhetoric-dominant not because users grew stupider, but because the platform's visibility rules rewarded rhetoric and punished argument.

§7.2 The Industrialization of the Hook (2014–2018)

Around 2014, formats of the "watch a whole movie in X minutes" type appeared on the Chinese internet—compressing hours of film and television into two-to-five-minute commentary segments. This was not just a new content type; it trained a declining tolerance for consuming long content: when three minutes can "get through" a movie, the 120-minute original becomes unbearably slow.

Gloria Mark's attention-tracking data moved in sync: 2.5 minutes on average in 2004 → 75 seconds in 2012 → 47 seconds in 2016 (median 40 seconds). Chiossi et al. (2023), in a controlled experiment from LMU Munich published at the CHI conference on human–computer interaction (N = 60), found that after the prospective-memory task was interrupted by TikTok, participants' cue-detection accuracy fell from about 80% to about 49% (a relative drop of nearly 40%)—while the Twitter and YouTube conditions showed no significant effect. The "hook" passed from an ancient trick of rhetoric into an industrial assembly line: the efficacy of every exposure measured, iterated, selectively bred. Persuasion did not grow more abundant—it acquired instrumentation.

§7.3 Format Compression and Cognitive Rewiring

The short-video format (15–60 seconds) combined with For You algorithmic recommendation is associated with changes in specific cognitive measures. Yan et al. (2024) examined the correlations between short-video addiction tendency and neural and behavioral measures (48 recruited; 45 included in electroencephalography (EEG) analysis): addiction tendency was significantly negatively correlated with prefrontal theta power (reflecting executive control) ($r = -0.395$, $p = 0.007$) and with self-control capacity ($r = -0.320$, $p = 0.026$). The researchers accordingly hypothesize that prolonged short-video consumption may capture attention with minimal mental effort, preferentially engaging lower-order regions associated with emotional processing while suppressing higher-order regions responsible for self-control—this is correlational evidence and cannot establish that short-video use caused these changes.

This converges, at the neural level, with the variable-reward mechanism recorded in §3: each swipe presenting unknown new content forms an intermittent reinforcement arrangement centered on uncertainty. Reeve proposes an explanatory hypothesis: intermittent dopamine surges may strengthen circuits associated with instant gratification and weaken activity associated with delayed reward—the studies cited in this chapter (correlational EEG evidence) cannot determine whether this neural mechanism holds (Reeve, 2025). The endpoint of format compression is not merely "watching shorter"; it is that deep understanding requiring reflection and analysis is often relegated to a secondary position—it must "compete within an ecosystem designed for fleeting attention" (Reeve, 2025).

§7.4 From Logic to Rhetoric: The Ontological Turn of Media

Richard Lanham, in the 1993 *The Electronic Word: Democracy, Technology, and the Arts*, delivered a diagnosis still unsurpassed: print carries the qualities of a philosophical medium (deeply serial, symbolic, chain-of-logic), while the electronic screen inclines toward the rhetorical (shallowly parallel, performative, immediately persuasive, competing for attention)—"Technology is following the main 'operating system' disagreement in our time, not driving it" (Lanham, 1993). In the same book he had already defined rhetoric as "the economics of attention, the allocation of a scarce resource" (Lanham, 1993). The 2006 *The Economics of Attention* develops the theme: when attention displaces matter as the scarce resource, style and substance swap places—style moves from margin to center, because capturing attention is style's entire function (Lanham, 2006).

Sine N. Just (2024), in *Controversial Encounters in the Age of Algorithms*, carries the line to the present: algorithms are "inherently rhetorical; they are shaped by and give shape to distinct forms of persuasion" (Just, 2024); citing Gillespie (2014), she notes that algorithms rest on "presumptions about what knowledge is and how one should identify its most relevant components." Public debate is squeezed from both sides—polarization (opposing groups turning antagonistic) and personalization (messages tailored to individual tastes)—resonating together into what she calls "the closing of the rhetorical mind": citizens increasingly unable to listen effectively to differing views and to produce persuasive messages (Just, 2024). The closure comes not because speech is censored—because the cognitive capacity for processing differing views has atrophied.

The format shaping recorded in §7.1–§7.3—from the internalization of 140 characters to the industrialization of hooks to cognitive rewiring—is the material basis of this atrophy. The platform does not forbid logic—it merely lets rhetoric tend to win the engagement contest. By the end of 2022, the most active expressers of a generation had been shaped into: hook openings, emotional closings, rhetorical fragments standing in for chains of argument. This is not yet "replacing you"—that is the next chapter's business—but it makes replacement easy: once the habit of complete argument has been persistently weakened, artificial intelligence (AI) saying "I'll write it for you" meets less resistance.

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§8 By 2022: What Had Been Trained (Handing Over to Chapter 28)

The preceding seven sections each recorded one stage in the building of an architecture. Now the view shifts from architecture to its product: by the end of 2022, what had users become?

Gloria Mark, professor of informatics at UC Irvine, has tracked screen attention since 2004 with stopwatches and computer logs in real work and life settings. Her long-run measurements show a clear descending curve: in 2004,

users switched away from any given screen after an average of about 150 seconds (2.5 minutes); by 2012, 75 seconds; from 2016 to 2021, a stable roughly 47 seconds (median 40 seconds—half of all observations below 40). Microsoft Research's independent measurement of 20 software developers found 50 seconds; Mark's doctoral student Akbar's tracking of 50 office workers found 44—the result was independently replicated (Mark, 2023). That is, by the end of the period this chapter covers, in her sample of information workers, screen attention span on workdays had fallen to about 47 seconds on average—and Chapter 7 has already noted that, given the heterogeneity of measurement methods and samples, this figure should be treated as suggestive rather than confirmatory evidence.

The figure moved in step with changes in the media environment. Mark records that the average shot length of film fell from about 12 seconds in 1930 and of television from about 13 seconds in 1950, converging below 4 seconds by 2010 (Mark, 2023, citing Cutting et al. and Butler). Facebook officially recommended to advertisers that mobile video ads not exceed 15 seconds, explicitly on the ground that user attention had shrunk to that length; 6-second ads had become a standard format (Mark, 2023). Algorithmic ranking, variable rewards, watch-time optimization—the product decisions recorded in §2–§4—did not operate on a stable user state but formed a positive feedback with the collapse of user attention: the shorter the attention, the more the algorithm depends on ultra-short formats to capture it; the more prevalent the ultra-short format, the shorter attention falls further. Mark describes this as a loop: our attention is shrinking while culture both adapts to and manufactures the conditions that keep it short (Mark, 2023).

At the aggregate level, global surveys (internet users aged 16–64) show that in 2022 average daily social-media time worldwide was 147 minutes—up 63% from 90 minutes in 2012 (Statista, 2025).

Placing these data on this chapter's narrative line, the state at the end of 2022 can be summarized in three layers. The first is the discretization of expressive format. From the long-form deep reading of the blog era (user-driven, high decision threshold), to the 140-character fragments of the microblog era (social-relation-driven, medium threshold), to the 15-second audiovisual flow of the short-video era (algorithm-driven, threshold approaching zero), the unit of content was continuously compressed. Each compression transferred control: from actively clicking links, to swiping through feeds, to "voting" purely by dwelling or swiping—initiative drained away step by step. The second is the complete transfer of distribution power. The material conditions recorded in §1 (price collapse, handset spread) supplied the data; the product decisions recorded in §2–§4 (algorithmic ranking, variable rewards, the interest graph) converted that data into the power to predict individual behavior, and that predictive power in turn determined what each user saw. By 2022, users no longer "looked for content"; content found them precisely. The third is the structural lag of governance. §6 recorded that Western governance responses had never been demonstrated to have changed the underlying architecture (China's new rules placed the recommendation mechanism under regulation earlier, with enforcement evidence pointing both ways): the law drew lines at data collection but found no grip on content distribution. In the major Western jurisdictions, the engagement-optimization architecture only became a direct object of rules in force with the DSA and DMA—and that gap only began to close (see §6).

This, then, is the cross-section at the end of 2022: a system capable of predicting individual behavior and accordingly rearranging its perceived world had been fully built and was running at every layer. It occupies roughly 147 minutes of surveyed internet users' (aged 16–64) daily social-media time; it is locked onto the fragmentation of users' screen attention—an average of about 47 seconds between screen switches in information-worker samples; it converts creators into functions of metrics, signaled by watch-through, rewind, and share rates; and

it shares almost none of its advertising revenue with content producers—on four major platforms the take rate runs near 99%, with YouTube's 45% the exception (see §4.3). Up to this point, what the system did was this: predict ever better what the user was about to do, and accordingly rearrange everything the user was about to see.

Placing this cross-section back within the book's coordinates: what the "Understanding You" phase completed was an upgrade in the Clock Room from recording to prediction—readings were no longer merely accumulated, but continuously converted back into rearrangements of the clock faces; the pseudo-fibers of Chapter 4 here expanded from individual perception to platform-scale manufacture (the more accurate the prediction, the more the user's perceived real was narrowed to the click stream itself); and, keeping accounts with Chapter 3's Macro-Cognitive Accounting Framework (MCAF), the narrative load (ϕ) attained over the decade ever-deepening leverage over the legitimacy base ($\|L\|$)—the power to rank, expanded without authorization. The keynote of "Understanding You" closes here: it is not that the platform understood desire; it is that the platform substituted predictability for the expression of desire.

The next step is no longer this. What Chapter 28 will record is this: the system no longer merely predicts what users want to see and ranks existing content—it generates the content itself. The user shifts from "being recommended" to "being generated"—search results are no longer a list of links but a direct answer; the interlocutor is no longer a person but the system itself; an output need not be preceded by a corresponding human work, but comes directly out of the model. This is not an extension of "Understanding You" but a new capability level: from predicting and ranking to generating and replacing. The architecture this chapter has recorded is that new capability's operational and distributional precondition—the behavioral data accumulated over a decade, the training and deployment infrastructure, and user habits compressed into ultra-short formats allow generated content to plug directly into a ready-made attention-allocation system; but generative models themselves do not depend on recommendation systems to be trained and run. Chapter 28 will take up from this cross-section.

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